

EXPERIMENT 8

NMAP LIVE HOST DISCOVERY

Discovering live / active hosts in a network

Using ARP, ICMP, TCP and UDP based discovery methods

1. AIM, OBJECTIVE & SOFTWARE

Aim

To discover live/active hosts in a network using Nmap. Scans such as ARP, ICMP, TCP and UDP are used.

Objective

To understand how Nmap identifies active systems before performing further network scanning.

Software / Requirement

- Nmap
- TryHackMe Nmap room / AttackBox
- Computer with network access

2. THEORY – NMAP HOST DISCOVERY

Nmap is a free and open-source network scanning tool used to discover live hosts and identify network services.

Nmap can discover live hosts using different protocols:

1. ARP Scan – Uses ARP scan requests to find hosts on the same subnet.
2. ICMP Scan – Uses ICMP requests and replies to identify active hosts.
3. TCP Scan – Uses TCP requests and packets to specified ports and checks the response.
4. UDP Scan – Sends UDP packets and can identify an active host through an ICMP port-unreachable response.

Key point: ARP discovery is limited to the local subnet because ARP operates at the link layer.

3. OBSERVATION & RESULT

Observation

- ARP requests can discover hosts within the same subnet.
- ARP cannot cross routers because it operates at the link layer.
- An ACK ping generally requires a privileged account.
- UDP ping can identify an active host when an ICMP port-unreachable response is received.
- The -sn option allows Nmap to perform host discovery without conducting a port scan.

Result

Thus, the live host discovery using Nmap was successfully studied and performed. The experiment helped in understanding how active hosts can be identified before performing further network scans.

EXPERIMENT COMPLETED SUCCESSFULLY